

CSE-302: Assignment-1

Requirement Specification Report

College Council & Club Management System

Group 4

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1 Introduction and Problem Statement

Within the college ecosystem, multiple councils govern a wide array of clubs. Each club may further divide into specialized verticals based on distinct operational or technical needs. Students frequently interact with these entities by applying for memberships, assuming leadership roles (PORs), and participating in various events. Furthermore, clubs require physical infrastructure, necessitating room allocations across various college hostels.

The current methodology requires students to fill out disparate forms for every single interaction. This decentralized approach leads to severe data redundancy, difficulty in tracking historical records of members and past events, and significant administrative overhead when allocating resources or verifying student credentials.

2 Detailed System Requirements

To effectively solve these issues, the proposed system must satisfy several functional and non-functional requirements:

2.1 Functional Requirements

- **Centralized Student & Role Management:** The system must maintain a unified directory of all students, tracking their current active club memberships, their specific verticals, and any Positions of Responsibility (POR) they hold.
- **Historical Archival:** The database must naturally separate and store data by academic year/tenure (e.g., 2025-2026), allowing administrators to view past members and past event results without manual data migration.
- **Automated Resource Allocation:** The system must handle the complex many-to-many relationship of room allocations, enabling clubs to access one or multiple rooms across different hostels, and allowing rooms to be shared if necessary.
- **Dynamic Application Workflows:** The system must support distinct, tailored forms for different request types (Membership Drives, Event Registrations, and Room Access), handling status updates (Pending, Approved, Rejected) smoothly.

2.2 Non-Functional Requirements

- **Scalability:** The database schema must be flexible enough to accommodate the addition of new councils, clubs, or entirely new types of events and applications in future academic years.
- **Data Integrity:** Strict relational constraints (Primary Keys and Foreign Keys) must be enforced to ensure no orphaned records exist (e.g., an event cannot be created for a non-existent club).

3 Proposed System Architecture

The platform will utilize a standard client-server architecture, emphasizing a strong relational database foundation.

3.1 Technology Stack

- **Database : MySQL.** A relational database is strictly required to enforce schema constraints, handle complex JOINS for multi-layered hierarchical data, and maintain ACID compliance during application form processing.
- **Frontend Application:** Web development technologies to create responsive, accessible portals for students to submit forms and for admins to review them.
- **Backend/API:** A web server responsible for executing business logic, securely querying the MySQL database, and returning data to the frontend client.

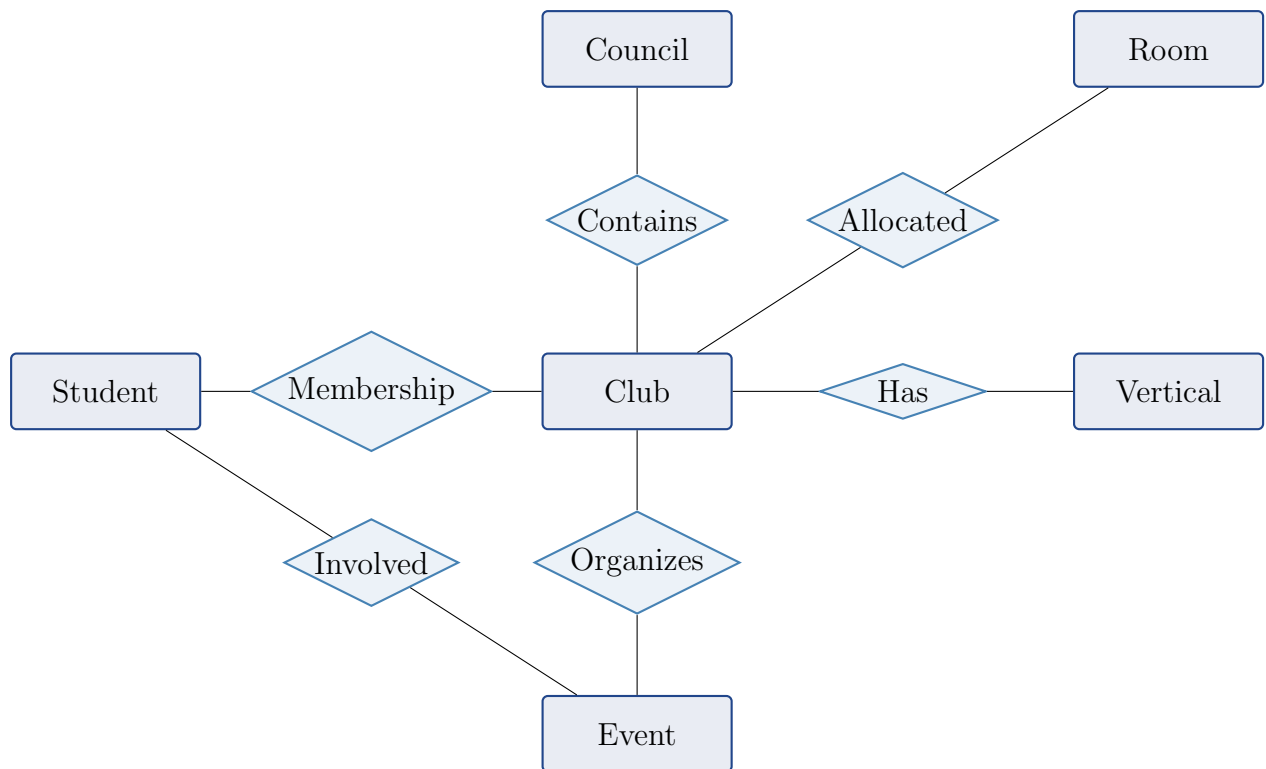
4 Entity-Relationship (ER) Conceptual Diagram

The foundation of the platform relies on a normalized relational schema. The core entities represent the hierarchical structure of the college organizations (*Council* $\rightarrow$ *Club* $\rightarrow$ *Vertical*) and the resources they interact with (*Student*, *Room*, *Event*).

4.1 Diagram Overview

The following diagram provides a high-level conceptual view.

- **One-to-Many (1:N):** A Council contains many Clubs. A Club has many Verticals and organizes many Events.
- **Many-to-Many (M:N):** Students participate in many events and are members of many clubs. Clubs can be allocated multiple rooms, and rooms might host multiple clubs. These M:N relationships are resolved in the schema using mapping tables.



Note: Form entities (Membership, Room Access, Event Reg) act as transactional requests pointing to these core entities and are omitted from this high-level diagram to maintain visual clarity.

5 Database Schema Definitions (MySQL)

The database is normalized to eliminate redundancy. Intermediary tables track academic-year-specific data. Transactional form data is separated into distinct tables to accommodate varying fields without sparse NULL columns.

5.1 Core Master Entities

These tables store the relatively static data about the college infrastructure and student body.

Table: Student	Data Type	Description
Roll_Number (PK)	VARCHAR(20)	Unique student identifier.
Name	VARCHAR(100)	Full name of the student.
Graduation_Year	INT	Expected year of graduation.
Branch	VARCHAR(50)	Department (e.g., Computer Science).
Email	VARCHAR(100)	Official college email ID.

Table: Council	Data Type	Description
Council_ID (PK)	INT	Unique identifier auto-increment.
Name	VARCHAR(100)	e.g., Cultural Council, Science & Tech.
Description	TEXT	Overview of the council's mandate.

Table: Club	Data Type	Description
Club_ID (PK)	INT	Unique identifier auto-increment.
Name	VARCHAR(100)	e.g., Fine Arts Club, Programming Club.
Council_ID (FK)	INT	Reference to parent Council.

Table: Vertical	Data Type	Description
Vertical_ID (PK)	INT	Unique identifier auto-increment.
Name	VARCHAR(50)	e.g., Charcoal Sketching, Frontend, Operations.
Club_ID (FK)	INT	Reference to parent Club.

Table: Event	Data Type	Description
Event_ID (PK)	INT	Unique identifier auto-increment.
Name	VARCHAR(150)	Title of the event or competition.
Club_ID (FK)	INT	Reference to the Club organizing it.
Event_Date	DATE	Scheduled date of the event.
Academic_Year	VARCHAR(10)	Academic session (e.g., "2025-2026").

5.2 Relational Mapping & History Tables

These tables map the relationships between core entities and use the `Academic_Year` attribute to maintain historical logs natively.

Table: Club_Membership	Data Type	Description
Membership_ID (PK)	INT	Auto-increment identifier.
Roll_Number (FK)	VARCHAR(20)	The student holding the membership.
Club_ID (FK)	INT	The club they belong to.
Vertical_ID (FK)	INT (NULL)	Specific vertical, if applicable.
POR_Title	VARCHAR(50)	e.g., Member, Secretary, Joint Secretary.
Academic_Year	VARCHAR(10)	Tracks tenure (e.g., "2026-2027").

Table: Club_Room_Allocation	Data Type	Description
Allocation_ID (PK)	INT	Auto-increment identifier.
Club_ID (FK)	INT	The club assigned to the space.
Room_ID (FK)	INT	The physical room allocated.
Academic_Year	VARCHAR(10)	Year of valid allocation.

Table: Event_Involvement	Data Type	Description
Involvement_ID (PK)	INT	Auto-increment identifier.
Roll_Number (FK)	VARCHAR(20)	The student involved.
Event_ID (FK)	INT	The target event.
Role_Category	VARCHAR(50)	e.g., 'Participant', 'Management', 'Volunteer'.
Specific_Role	VARCHAR(50)	e.g., 'Event Coordinator' (POR), 'Contestant'.
Result_Details	TEXT	(Nullable) e.g., '1st Place', feedback, or task details.

5.3 Application Forms (Transactional)

These tables manage the automated workflows. Upon a status changing to 'Approved', backend logic will automatically generate records in the respective mapping tables (e.g., an approved `Form_Membership` triggers an insert into `Club_Membership`).

Table: Form_Membership	Data Type	Description
Form_ID (PK)	INT	Unique application ID.
Roll_Number (FK)	VARCHAR(20)	Applicant.
Club_ID (FK)	INT	Target club.
Vertical_ID (FK)	INT (NULL)	Target vertical.
Statement_Of_Purpose	TEXT	Applicant's motivation / qualifications.
Status	VARCHAR(20)	'Pending', 'Approved', 'Rejected'.
Academic_Year	VARCHAR(10)	Session applied for.

Table: Form_EventReg	Data Type	Description
Form_ID (PK)	INT	Unique application ID.
Roll_Number (FK)	VARCHAR(20)	Applicant.
Event_ID (FK)	INT	Target event.
Applied_Category	VARCHAR(50)	e.g., applying for 'Management' vs 'Participation'.
Team_Name	VARCHAR(100)	(Nullable) If participating as a team.
Status	VARCHAR(20)	'Pending', 'Approved', 'Rejected'.

Table: Form_RoomAccess	Data Type	Description
Form_ID (PK)	INT	Unique application ID.
Roll_Number (FK)	VARCHAR(20)	Applicant requesting access.
Room_ID (FK)	INT	Target room.
Reason	TEXT	Justification for access.
Status	VARCHAR(20)	'Pending', 'Approved', 'Rejected'.

6 Conclusion

By transitioning from isolated, manual forms to a centralized, relational MySQL database-driven application, the College Council & Club Management System will drastically reduce administrative friction. This robust backend architecture, paired with an accessible web frontend, provides a scalable foundation that can seamlessly adapt to evolving institutional requirements and expanding club structures in the future.